Newton solver with Armijo line search using TinyAD

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Chapter 1

Todo List

Member NewtonSolver< N, M >::set_settings (int max_iterations, double tol)

Also add possibility of setting Armijo line search parameters.

Chapter 2

Class Index

2.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Chapter 3

File Index

3.1 File List

Here is a list of all documented files with brief descriptions:

Chapter 4

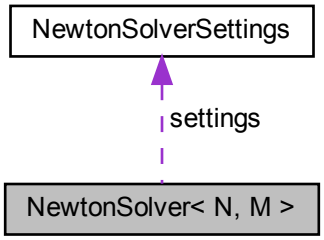
Class Documentation

4.1 NewtonSolver< N, M > Class Template Reference

Newton solver for systems of nonlinear equations.

`#include <Newton_solver.hh>`

Collaboration diagram for NewtonSolver< N, M >:



Public Types

- using **AD_N** = TinyAD::Double< N >
- using FuncSig = void(*)(const Eigen::Matrix< AD_N, N, 1 > &, Eigen::Matrix< AD_N, N, 1 > &)

  It is the signature of the function that computes the system of equations to be differentiated. It takes as input an Eigen vector of AD scalars, and outputs an Eigen vector of AD scalars, which is the evaluation of the system of equations at the input point. The user has to provide a function with this signature to compute the system of equations to be solved.
- using FuncSigDouble = void(*)(const Eigen::Matrix< double, N, 1 > &, Eigen::Matrix< double, N, 1 > &)

  Signature for the same function asa bove, but without automatic differentiation. Used when just an evaluation of the function is needed.

Public Member Functions

- bool get_success () const

  Returns whether the solver converged successfully.
- int get_iterations () const

  Returns the number of iterations performed by the solver.
- void set_settings (int max_iterations, double tol)

  Returns the number of iterations performed by the solver.
- NewtonSolverSettings get_settings () const

  Returns the settings struct.
- void set_initial_guess (Eigen::Matrix< double, N, 1 > &guess)

  Provide an initial guess to the solver.
- void **print_initial_guess** () const
- void Jacobian (const Eigen::Matrix< AD_N, N, 1 > &input, FuncSig func)

  Function to compute Jacobian matrix with automatic differentiation.
- const Eigen::Ref< const Eigen::MatrixXd > get_solution () const

  Returns the solution found by the solver.
- void solve (FuncSig func, FuncSigDouble func_double)

  Solve the problem, given the function to be solved, coherent in size with the instantiated solver.

Protected Member Functions

- void assign_F (const Eigen::Matrix< AD_N, M, 1 > &F_active)

  Function to store the value of an eigen vector of AD scalars into the attribute F_val that stores the value of the system of equations.
- template<int Size>
  void assign_AD_vector (const Eigen::Matrix< AD_N, Size, 1 > &vec_source, Eigen::Matrix< double, Size, 1 > *&vec_dest)

  Function to store the value of an eigen vector of AD scalars into an eigen vector of doubles. It is templetized on the size of the vectors.
- void assign_solution (const Eigen::Matrix< AD_N, N, 1 > &sol)

  Function to store the value of an eigen vector of AD scalars into the solution attribute, which is an eigen vector of doubles.
- double merit_fun (const Eigen::Matrix< double, N, 1 > &x, Eigen::Matrix< double, M, 1 > *&tmp, FuncSigDouble func)

  *Computes the merit function value: phi(x) = 0.5*norm_2(F(x))^2.*
- double armijo_search (Eigen::Matrix< double, N, 1 > &x, Eigen::Vector< double, N > &d, FuncSigDouble func, Eigen::Vector< double, N > &f_val_x)

  Performs Armijo line search to find a suitable step size t that satisfies the Armijo condition.

Protected Attributes

- NewtonSolverSettings **settings**
- bool **success** = false
- int **iterations** = 0
- Eigen::Vector< double, N > **solution** = Eigen::Matrix<double, N, 1>::Zero()
- Eigen::Matrix< double, N, 1 > * **initial_guess** = new Eigen::Matrix<double, N, 1>(Eigen::Matrix<double, N, 1>::Zero())
- Eigen::Matrix< double, M, N > * **J** = nullptr
- Eigen::Matrix< double, M, 1 > * **F_val** = nullptr
- Eigen::Matrix< AD_N, M, 1 > * **F_val_ad** = nullptr
- Eigen::Matrix< double, N, 1 > **d** = Eigen::Matrix<double, N, 1>::Zero()
- Eigen::Matrix< double, M, 1 > * **lhs_armijo** = nullptr
- Eigen::Matrix< double, M, 1 > * **rhs_armijo** = nullptr

4.1.1 Detailed Description

template<int N, int M>
class NewtonSolver< N, M >

Newton solver for systems of nonlinear equations.

It is a class that exploits automatic differentiation to compute the Jacobian matrix of the system of equations at each iteration, and uses it to compute the descent direction, employing Armijo line search. It is templatized on the number of variables and equations, and it require the user to provide the function that computes the system of equations to be differentiated.

Template Parameters

N	number of variables
M	number of equations

4.1.2 Member Function Documentation

4.1.2.1 armijo_search()

```
template<int N, int M>
double NewtonSolver< N, M >::armijo_search (
            Eigen::Matrix< double, N, 1 > & x,
            Eigen::Vector< double, N > & d,
            FuncSigDouble func,
            Eigen::Vector< double, N > & f_val_x )  [inline], [protected]
```

Performs Armijo line search to find a suitable step size t that satisfies the Armijo condition.

Parameters

x	The current point in the optimization.
d	The descent direction found witht he Jacobian and the resolution of the LU system.
func	The function for which to perform the line search.
f_val_x	The value of the function at the current point x, used to compute the Armijo condition.

Returns

The step size t that satisfies the Armijo condition.

4.1.2.2 assign_AD_vector()

```
template<int N, int M>
template<int Size>
```

```
void NewtonSolver< N, M >::assign_AD_vector (
            const Eigen::Matrix< AD_N, Size, 1 > & vec_source,
            Eigen::Matrix< double, Size, 1 > *& vec_dest )  [inline], [protected]
```

Function to store the value of an eigen vector of AD scalars into an eigen vector of doubles. It is templetized on the size of the vectors.

Parameters

vec_source	The eigen vector of AD scalars to be stored in vec_dest.
vec_dest	The eigen vector of doubles where the values will be stored.

4.1.2.3 assign_F()

```
template<int N, int M>
void NewtonSolver< N, M >::assign_F (
            const Eigen::Matrix< AD_N, M, 1 > & F_active )  [inline], [protected]
```

Function to store the value of an eigen vector of AD scalars into the attribute F_val that stores the value of the system of equations.

Parameters

F_active	The eigen vector of AD scalars to be stored in F_val.

4.1.2.4 get_iterations()

```
template<int N, int M>
int NewtonSolver< N, M >::get_iterations ( ) const  [inline]
```

Returns the number of iterations performed by the solver.

Returns

The number of iterations performed.

4.1.2.5 get_success()

```
template<int N, int M>
bool NewtonSolver< N, M >::get_success ( ) const  [inline]
```

Returns whether the solver converged successfully.

Returns

true if the solver converged, false otherwise.

4.1.2.6 Jacobian()

```
template<int N, int M>
void NewtonSolver< N, M >::Jacobian (
            const Eigen::Matrix< AD_N, N, 1 > & input,
            FuncSig func )  [inline]
```

Function to compute Jacobian matrix with automatic differentiation.

It does not have return but it assigns the values of the Jacobian to the J attribute. It also keep track of the value of the function at the current input point, which is stored in F_val_ad.

Parameters

input	The point at which to compute the Jacobian, it must be an eigen vector of AD scalars.
func	The function that computes the system of equations. It must have the signature FuncSig.

4.1.2.7 merit_fun()

```
template<int N, int M>
double NewtonSolver< N, M >::merit_fun (
            const Eigen::Matrix< double, N, 1 > & x,
            Eigen::Matrix< double, M, 1 > *& tmp,
            FuncSigDouble func )  [inline], [protected]
```

Computes the merit function value: phi(x) = 0.5∗norm_2(F(x))^2.

Parameters

x	The point at which to evaluate the merit function.
tmp	Temporary vector for intermediate computations.
func	The function for which to compute the merit function.

Returns

The value of the merit function.

4.1.2.8 set_settings()

```
template<int N, int M>
void NewtonSolver< N, M >::set_settings (
            int max_iterations,
            double tol )  [inline]
```

Returns the number of iterations performed by the solver.

Returns

The number of iterations performed.

Todo Also add possibility of setting Armijo line search parameters.

The documentation for this class was generated from the following file:

- include/Newton_solver.hh

4.2 NewtonSolverSettings Struct Reference

`#include <Newton_solver.hh>`

Public Attributes

- int max_iterations = 1000
- double tol = 1e-9
- double gamma = 1e-4
- double beta = 0.5
- double t_max = 1.0

4.2.1 Detailed Description

Simple struct to hold solver settings

4.2.2 Member Data Documentation

4.2.2.1 beta

`double NewtonSolverSettings::beta = 0.5`

Beta for Armijo line search, parameters that multiplies t to update it at each iteration during line search

4.2.2.2 gamma

`double NewtonSolverSettings::gamma = 1e-4`

Gamma for Armijo line search

4.2.2.3 max_iterations

```
int NewtonSolverSettings::max_iterations = 1000
```

Maximum number of iterations

4.2.2.4 t_max

```
double NewtonSolverSettings::t_max = 1.0
```

Maximum step size and initial value for Armijo line search

4.2.2.5 tol

```
double NewtonSolverSettings::tol = 1e-9
```

Tolerance for convergence

The documentation for this struct was generated from the following file:

- include/Newton_solver.hh

Chapter 5

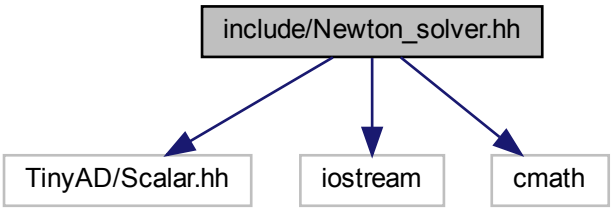
File Documentation

5.1 include/Newton_solver.hh File Reference

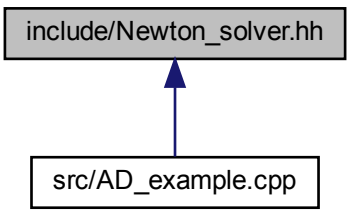
Newton solver with Armijo line search using TinyAD.

```
#include <TinyAD/Scalar.hh>
#include <iostream>
#include <cmath>
```

Include dependency graph for Newton_solver.hh:



This graph shows which files directly or indirectly include this file:



Classes

- struct NewtonSolverSettings
- class NewtonSolver< N, M >

  Newton solver for systems of nonlinear equations.

5.1.1 Detailed Description

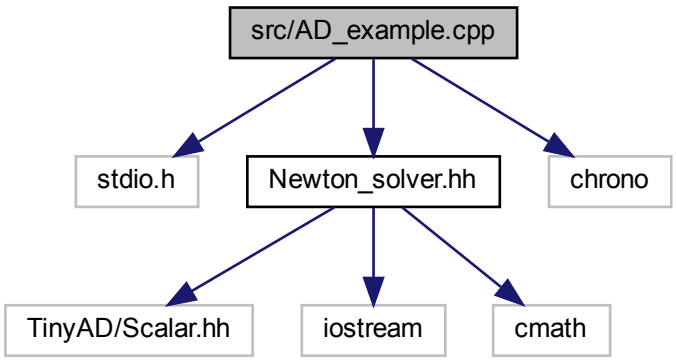
Newton solver with Armijo line search using TinyAD.

5.2 src/AD_example.cpp File Reference

Example of root finding for nonlinear systems with the Newton solver.

```
#include <stdio.h>
#include "Newton_solver.hh"
#include <chrono>
```

Include dependency graph for AD_example.cpp:



Functions

- template<typename Scalar >
  void HelicalValley (const Eigen::Matrix< Scalar, 3, 1 > &v, Eigen::Matrix< Scalar, 3, 1 > &out)

  Helical Valley function, a common test problem for optimization algorithms. Every systems of equation has to be described by a function of this form, that is passed as argument to the solver.
- int **main** ()

5.2.1 Detailed Description

Example of root finding for nonlinear systems with the Newton solver.

5.2.2 Function Documentation

5.2.2.1 HelicalValley()

```
template<typename Scalar >
void HelicalValley (
            const Eigen::Matrix< Scalar, 3, 1 > & v,
            Eigen::Matrix< Scalar, 3, 1 > & out )
```

Helical Valley function, a common test problem for optimization algorithms. Every systems of equation has to be described by a function of this form, that is passed as argument to the solver.

Template Parameters

Scalar	The scalar type of the input and output vectors. The solver requires double and double AD scalar type, the type used by TinyAD.

Parameters

v	The input vector representing the point at which to evaluate the function.
out	The output vector where the function value will be stored.

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